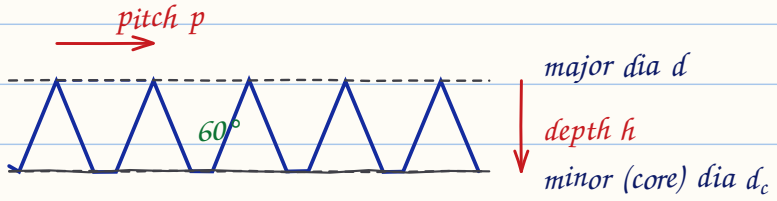
UNIT - II

Design of Bolted and Welded Joints

1. Threaded Fasteners

Detachable joints - bolts, screws, studs, nuts - clamp parts together by tension in the shank created by tightening the nut. Threads are an inclined plane (helix) wrapped on a cylinder.



ISO metric thread - V form, 60°

Fig 1: Screw thread terminology

- Pitch p : axial distance between adjacent threads ; lead $L = n\ p$ (n = number of starts) ; single start → lead = pitch.
- Major dia d (nominal) , minor / core dia d_c , pitch dia $d_p = (d + d_c)/2$; helix angle α : $\tan \alpha = L / \pi d_p$.
- Tensile stress area $A_t = (\pi/4)(d - 0.9382\ p)^2 \approx (\pi/4)\ d_c^2$ with $d_c \approx 0.8\ d$. Always use A_t (or core area) for stress in the bolt.
- Metric designation : M20 × 2.5 → nominal dia 20 mm, coarse pitch 2.5 ; M20 × 1.5 → fine thread (better for vibration, shallow depth, bigger core dia, adjustment).

Thread form	Features	Use
ISO metric (V, 60°)	high friction, strong core, self-locking	bolts, nuts, screws
Square	efficient, no radial thrust, difficult to cut	power screws, jacks
Acme (29°)	easy to make, wear adjustable by split nut	lead screws, vices
Buttress	strong in one direction only	press, jack, vice
Knuckle / round	rounded, resists damage, dirt	bulbs, hose couplings

Common fasteners & locking devices

- Through bolt (nut at the other end) , tap bolt (screws into a tapped hole, nut not used) , stud (threaded both ends, for cylinder heads) , cap screw , set screw (hold hub on shaft).
- Locking devices against vibration loosening : jam (lock) nut, castle nut + split pin, spring (Grover) washer, toothed lock washer, Sarcotite / nylon insert (Nylock) nut, wire locking, tab washer, set screw.

Bolt grades (property class)

Class a.b : S_{ut} = 100 a MPa ; S_y = (b/10) S_{ut}.

Class	4.6	5.8	8.8	10.9	12.9
S_{ut} (MPa)	400	500	800	1000	1200
S_y (MPa)	240	400	640	900	1080

2. Preload & Stresses in Bolts

Bolts are always tightened to an initial tension (preload) F_i. Preload keeps the joint compressed, so that an external load does not open it, increases fatigue life (bolt sees only a part of the load), and prevents leakage / loosening.

$$F_i = 2840\ d\ N \quad (d \text{ in mm ; Bhandari - reusable joints})$$

$$F_i = 0.75\ A_t\ S_p \text{ (reusable)} , \quad 0.90\ A_t\ S_p \text{ (permanent)}$$

S_p = proof strength ≈ 0.85 S_y (class 8.8 : 600 MPa ; 10.9 : 830 MPa).

Various stresses induced in bolts

Cause	Stress	Formula
Initial tightening	direct tension	$\sigma_t = F_i / A_t$
Torque while tightening	torsional shear	$\tau = 16\ T_1 / \pi\ d_c^3$ ($T_1 \approx$ ½ T)
External axial load P	additional tension	$\sigma = (F_i + C\ P)/A_t$
Transverse (shear) load	direct shear	$\tau = F_s/(\pi/4\ d^2)$, bearing $\sigma_c = F_s/(d\ t)$
Thread contact	shear of threads	$\tau = F / (\pi\ d_c \cdot 0.5\ p \cdot z)$; nut H ≈ 0.8d safe
Non-parallel seats	bending	$\sigma_b = E\ d\ \theta\ /\ 2L$ - avoid by spot facing

Note: Because tightening twists the bolt, design on the combined stress : bolt is made ~30% stronger : $A_t \geq 1.3\ F_i / \sigma_{allow}$ (or use von Mises with σ, τ from torque).

Design of a bolt for static loads

- Axial pull P : $A_t = P / \sigma_{allow}$, $\sigma_{allow} = S_y/n$ (n = 2-3 for static) → choose standard M-size from the table of core areas (M10 : 58 mm² , M12 : 84 , M16 : 157 , M20 : 245 , M24 : 353 mm²).
- Transverse (shear) load : turned / fitted bolt in reamed hole : $\tau = 4\ F_s / (\pi\ d^2) \leq S_{sy}/n$; crushing $\sigma_c = F_s/(d\ t)$. Ordinary bolt in clearance hole : load is carried by friction, $\mu\ F_i\ i \geq F_s$ ($\mu \approx 0.15$-0.2, i = number of interfaces) → large preload needed.
- Bolts of uniform strength : to improve shock / fatigue strength drill axial hole in the shank or reduce shank dia to core dia (d_c) so that the strain energy stored ($\frac{1}{2}\ \sigma^2\ A\ L / E$) is spread evenly and thread-root stress concentration is not the weak point.
- Nut height : $H = 0.8\ d$ standard ; thread-stripping shear area $\approx \pi\ d_c\ (0.5\ H)$ per nut ; nut is made softer (grade ≈ one lower than bolt) so load spreads over the first 3 threads - first thread carries ~ 30%, 3rd ~ 10%. Using a taller nut does not help beyond ~ 6 threads.

Example (axial) : bolt carries a steady pull of 20 kN, class 4.6 ($S_y = 240$), $n = 3 \rightarrow \sigma_{allow} = 80$ MPa. $A_t = 20000/80 = 250$ mm² → M20 ($A_t = 245 \approx$ OK) ; with torque factor 1.3 → 325 mm² → M24.

Torque requirement for tightening

$$T = F_i \left[(d_p/2)\left((\mu + \tan\alpha)/(1 - \mu\ \tan\alpha) \right) + \mu_c\ d_c/2 \right]$$

$$T = K\ F_i\ d \quad (K \approx 0.2 \text{ for dry / plain steel})$$

- First term : thread (screw) friction ; second : collar / nut face friction (mean collar dia d_c). K (torque coefficient) = 0.15 lubricated , 0.2 dry , 0.3 rough / plated.
- Efficiency of tightening is poor : ~ 10% of torque makes tension, rest is lost in friction → large scatter ±25% unless torque wrench or turn-of-nut method is used.

- Spanner length for hand tightening ≈ 15-20 d. Standard spanner on M16 → easily overloads M8 - avoid.

Screw mechanics - self-locking & efficiency

$$\tan \alpha = p / \pi d_p \quad , \quad \mu' = \mu / \cos 30° \ (V \text{ thread})$$

Self-locking if $\mu' > \tan \alpha$ (always true for fasteners, $\alpha \approx 2\text{-}3°$)

$$\eta = \tan \alpha \ (1 - \mu' \tan \alpha) / (\mu' + \tan \alpha) \approx 20\text{-}25\ \%$$

M20 × 2.5 : $d_p = 18.4$ mm , $\tan \alpha = 0.0433$; with $\mu' = 0.15 \rightarrow \eta = 22$ %. Low efficiency = good locking but 78% of the torque goes in friction ; lubrication changes F_i by ±30% for the same torque !

Example 1 : M16 bolt (d = 16 mm). $F_i = 2840 \times 16 = 45.4$ kN, $T = 0.2 \times 45.4 \times 10^3 \times 16 = 145$ N·m. Core area ≈ 144 mm² (M16×2) → $\sigma = 45400/144 = 315$ MPa → class 8.8 or higher required.

3. Bolted Joint with External Load - Joint Stiffness

When an external tensile load P (axial, pulls the plates apart) is applied to a preloaded joint, the bolt stretches more and the compressed plates relax. Bolt stiffness $k_b = A_b E / L$, member (plates + washers) stiffness k_m (conical pressure frustum model).

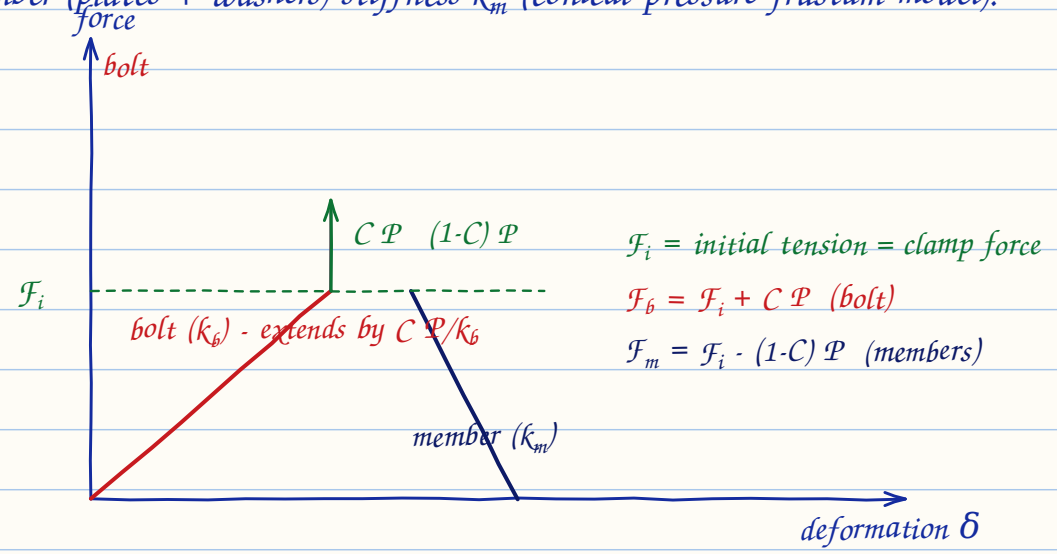


Fig 2: Load - deformation diagram of a preloaded joint

$$C = k_b / (k_b + k_m) \quad \text{(joint constant, 0.1 - 0.4)}$$

$$F_b = F_i + C P \quad , \quad F_m = F_i - (1 - C) P$$

- Separation of the joint (leak) when $F_m = 0 \rightarrow P_o = F_i / (1 - C)$. Design : $P < P_o$; factor of safety against separation $n_o = P_o/P$.
- Smaller C (stiff plates, flexible long bolt) → bolt takes only a small part of P → better fatigue life ; this is why long thin bolts, reduced shank and hard plates are good.
- Bolt must also not yield : $n_p = (S_p A_t - F_i) / C P$ (load factor).

Gasketed joints (pressure vessel covers, cylinder heads)

Total force on the cover $P_{total} = (\pi/4) D^2 p_{fluid}$. With z bolts on bolt-circle diameter D_b : $P = P_{total}/z$ per bolt. A soft gasket has small k_m → $C \approx 0.6$-0.9 (the bolt takes most of P) , a hard / metal gasket → $C \approx 0.2$-0.4.

- Gasket must retain a minimum seating pressure for no leak : $F_m = F_i - (1 - C)P \geq$ gasket seating load. Hence a high F_i ($\approx 0.9 A_t S_p$) is chosen in leak-proof joints.
- Number of bolts & pitch : bolt pitch = $\pi D_b/z$. Keep pitch ≥ 3d (spanner space) and ≤ $20\sqrt{t}$ (t = flange thickness, mm) so that the gasket pressure stays uniform. For high-pressure joints pitch ≈ 3d to 5d ; at least 4 bolts, usually even.

$$\text{Resultant load on a bolt : } P_{res} = P_i + C P \leq \sigma_{allow} A_t$$

Example 2 : cylinder cover, D = 300 mm, p = 1.5 MPa, 8 bolts. $P_{total} = (\pi/4)\, 300^2 \times 1.5 = 106$ kN → P = 13.25 kN per bolt. Take C = 0.3, $F_i = 2840$ d. With M16 : $F_i = 45.4$ kN. $F_b = 45.4 + 0.3 \times 13.25 = 49.4$ kN → $\sigma = 49400/144 = 343$ MPa < $0.85 S_y$ for class 8.8 (544). Safe ; also $F_m = 45.4 - 0.7 \times 13.25 = 36.1$ kN > 0 (no leakage).

Fatigue loading of bolts (fluctuating P)

$$\sigma_a = K_f C (P_{max} - P_{min}) / 2 A_t$$
$$\sigma_m = [\, F_i + C (P_{max} + P_{min})/2 \,] / A_t$$
$$\text{Goodman : } \sigma_a/S_e + \sigma_m/S_{ut} = 1/n$$

K_f (threads) ≈ 2.2 (rolled) to 3.8 (cut, hard) ; S_e for bolts (axial) ≈ $0.5 S_{ut} \times k_a$ … ~ 130-180 MPa for class 8.8 rolled threads. Higher F_i → σ_m higher but σ_a unchanged ; bolt life improved by preload up to proof limit.

Example 3a (fatigue of a cover bolt) : M12 ($A_t = 84$ mm²), class 8.8 ($S_{ut} = 800$), rolled threads $K_f = 3$, $S_e = 150$ MPa. P varies 0 → 6 kN per bolt, $F_i = 15$ kN, C = 0.25. $\sigma_a = 3 \times 0.25 \times 6000 / (2 \times 84) = 26.8$ MPa. $\sigma_m = (15000 + 0.25 \times 3000)/84 = 187.5$ MPa.

Goodman : $1/n = 26.8/150 + 187.5/800 = 0.413 \rightarrow n = 2.4$. Yield check : $F_{b,max} = 15000 + 0.25\times6000 = 16.5$ kN $\rightarrow 196$ MPa $< S_p = 600$ (OK).

Steps in bolted joint design

- (1) Find external load per bolt P (pressure × area / z) ; (2) choose C from gasket / plate stiffness ; (3) choose F_i (2840 d or 0.75-0.9 $A_t S_p$) ; (4) $F_b = F_i + C P \rightarrow A_t = 1.3 F_b / \sigma_{allow}$; (5) check separation $P_o = F_i/(1 - C) > P$; (6) check fatigue (Goodman) if load fluctuates ; (7) select standard size, torque $T = 0.2 F_i d$.

4. Eccentrically Loaded Bolted Joints

(a) Load in the plane perpendicular to bolt axes (bracket in shear)

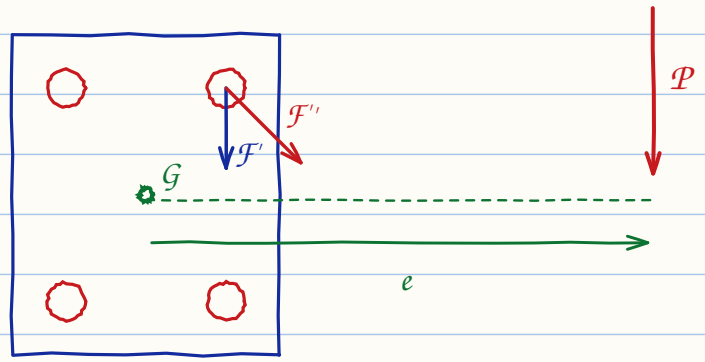


Fig 3: Eccentric load P on a bolt group (bolts in shear)

- Shift P to the centroid G of the bolt group : P (direct) + moment $M = P e$.
- Primary (direct) shear $F' = P / n$ on every bolt, parallel to P.
- Secondary shear $F''_i = M r_i / \Sigma r^2$ (same for equal bolts : $F'' = P e / (n r)$), perpendicular to the line joining the bolt to G.

$$F = \sqrt{(F'^2 + F''^2 + 2 F' F'' \cos\theta)}$$

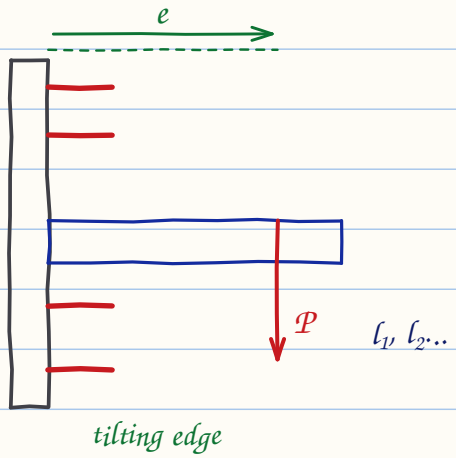
$$F'' \propto r \rightarrow F''_1/r_1 = F''_2/r_2 = \ldots$$

Bolt design : $\tau = F / (\pi/4\, d_c^2) \le S_{sy}/n$ ($S_{sy} = 0.5 S_y$) ; check bearing on the plate $\sigma_c = F/(d t) \le S_y/n$. If bolts are in clearance holes the load is carried by friction $\mu F_i n_s \ge P$ (slip-critical).

Example 3 : 4 bolts at corners of a 100 mm square ($r = 70.7$ mm), $P = 12$ kN, $e = 200$ mm. $F' = 3000$ N ; $F'' = 12000 \times 200 / (4 \times 70.7) = 8485$ N. For the worst bolts $\theta = 45°$: $F = \sqrt{(3000^2 + 8485^2 + 2\times3000\times8485\times0.707)} = 10.8$ kN. Class 4.6 ($S_y = 240$), $n =$

2.5 : $\tau_{allow} = 0.5 \times 240/2.5 = 48$ MPa → $d_c^2 = 4 \times 10820/(\pi \times 48) = 287$ → $d_c = 16.9$ mm → **M20** ($d_c = 17.3$).

(b) Load parallel to the bolt axes, eccentric (bracket in tension)



$F_i \propto$ distance l from the tilting edge

$P\,e = \Sigma\, F_i\, l_i = (F_1/l_1)\, \Sigma\, l_i^2$

Fig 4: Bracket tilting about the lower edge

Under P e the bracket tends to tilt about the edge of the base (assuming the plate rigid, the bolt elongation ∝ distance l from the tilting edge ; bolts near the edge carry little).

$$F_i \text{ (bolt at } l_i) = F_1\, l_i/l_1$$

$$P\,e = \Sigma\, F_i\, l_i \;\rightarrow\; F_1 \text{ (max)} = P\,e\, l_1/\Sigma\, l^2$$

$$\text{Direct load per bolt} = P/z \;\rightarrow\; \text{total } F_{max} = P/z + P\,e\, l_1/\Sigma\, l^2$$

- Initial tightening is superimposed ; if joint-compressive preload is large, the whole base remains in contact (the tilt-edge assumption is conservative). Design : $A_t = F_{max}/\sigma_{allow}$.

Example 6 : bracket with 4 bolts (2 rows × 2) in tension, $P = 20$ kN at $e = 200$ mm from the wall. Lower bolt row is the tilt-edge side : rows are at $l_1 = 150$ mm (upper, farthest) and $l_2 = 50$ mm. Per row 2 bolts. $\Sigma\, l^2 = 2\,(150^2 + 50^2) = 50000$ mm². $F_1 = P\,e\, l_1/\Sigma\, l^2 = 20000 \times 200 \times 150/50000 = 12$ kN per bolt. Direct load $P/4 = 5$ kN → $F_{max} = 17$ kN. Class 5.8 ($S_y = 400$) , $n = 2$ → $\sigma_{allow} = 200$ MPa → $A_t = 17000/200 = 85$ mm² → **M12** (84 mm², with preload fine - take M14).

- Combined case (load inclined / both shear + tension) : σ_t and τ on the same bolt - use equivalent stress $\sigma_{eq} = \sqrt{(\sigma_t^2 + 3\tau^2)}$ or maximum shear theory $\sqrt{(\sigma_t^2 + 4\tau^2)} \le S_y/n$.

5. Welded Joints

Permanent joint - parts fused with / without filler metal (arc, gas, resistance, TIG, MIG, submerged arc). Advantages : no hole weakening, lighter, cheaper, leak-proof, can join complex shapes. Disadvantages : residual stress & distortion, poor fatigue, skilled work needed, heat-affected zone.

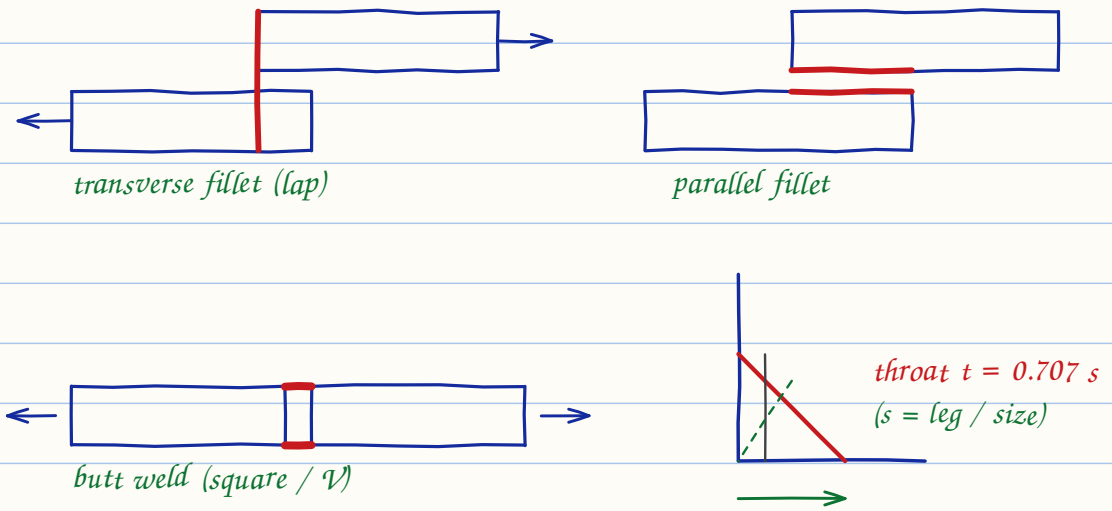


Fig 5: Types of welded joints, fillet throat

- Butt joint (square, single / double V, U, J, bevel) - plates in the same plane ; lap joint (fillet transverse or parallel) ; T joint , corner , edge ; plug / slot / spot welds.
- Design standards : IS 816 (arc welding of mild steel), AWS D1.1, IS 9595. Weld symbol : arrow + reference line + weld symbol (triangle = fillet, V = single V groove) + size s + length.

6. Strength of Butt and Fillet Welds

Butt weld : $P = t\, l\, \sigma_t$

(t = plate thickness = throat , l = length)

Transverse fillet (single) : $P = 0.707\, s\, l\, \sigma_t$

Parallel fillet (single) : $P = 0.707\, s\, l\, \tau$

Double parallel fillet : $P = 1.414\, s\, l\, \tau$

- Fillet weld cross-section is a right triangle of leg s ; failure plane through the throat $t = s \cos 45° = 0.707\, s$. For a transverse fillet the plane is at 67.5° and the stress is ~ τ (conservative to treat it like this).

- Allowable (static, IS 816) : butt in tension 90-110 MPa ; fillet in shear 80 MPa ; reduce by 0.8 (field weld) and for fatigue use K_f below. Weld strength = 1 for reinforced full-penetration butt weld.

Weld location	K_f (fatigue)
Reinforced butt weld	1.2
Toe of transverse fillet weld	1.5
End of parallel fillet weld	2.7
T-butt joint, sharp corner	2.0

- Combined lap joint (transverse l_1 + double parallel l_2) : $P = 0.707\ s\ l_1\ \tau + 1.414\ s\ l_2\ \tau$ (total effective length). Plug / spot weld : $P = (\pi/4)\ d^2\ \tau$, d = hole / spot diameter.
- Joint efficiency η_w (strength of joint / strength of plate) : full-penetration butt 0.9-1.0 ; double fillet lap ≈ 0.7-0.8 ; single fillet ≈ 0.55. Design load = η_w × plate strength.

Axially loaded unsymmetrical welds (angle section on a plate)

Line of action of P passes through the centroid G of the angle ; welds l_1 and l_2 at distances y_1, y_2 from G must have equal moment about G so that no eccentricity arises :

$$l_1\ y_1 = l_2\ y_2 \quad \text{and} \quad l_1 + l_2 = l$$
$$P = 0.707\ s\ (l_1 + l_2)\ \tau_{allow}$$

Example 4 : Plate 100 × 10 mm is welded to a frame by two parallel fillet welds (one each side) to carry $P = 80$ kN. $\tau_{allow} = 80$ MPa. $P = 1.414\ s\ l\ \tau \rightarrow s\ l = 80000 / (1.414 \times 80) = 707$ mm². Take $s = 8$ mm $\rightarrow l = 88.4$ mm ; add 2s for crater / end return → provide $l = 105$ mm on each side ($l \geq$ plate width 100 mm, OK).

7. Welds in Bending and Torsion

(a) Welds subjected to bending

Treat the weld group as a line of unit throat width. Then $\sigma = M / (0.707\ s\ Z_u)$, where Z_u is the section modulus of the weld lines (mm²).

Weld group (lines)	Z_u	J_u (torsion)
one vertical line, length d	$d^2 / 6$	$d^3 / 12$
two vertical lines (d long, b apart)	$d^2 / 3$	$d(3b^2 + d^2)/6$
two horizontal lines (b long, d apart)	$b\,d$	$b(3d^2 + b^2)/6$
rectangle all round ($b \times d$)	$d(3b + d)/3$	$(b + d)^3/6$
circular weld (dia d)	$\pi\, d^2 / 4$	$\pi\, d^3 / 4$

Transverse shear V adds $\tau' = V / (0.707\, s\, L)$, L = total weld length ; resultant $\tau = \sqrt{(\tau'^2 + \tau_b^2)}$.

(b) Eccentric load in the plane of the weld (torsion)

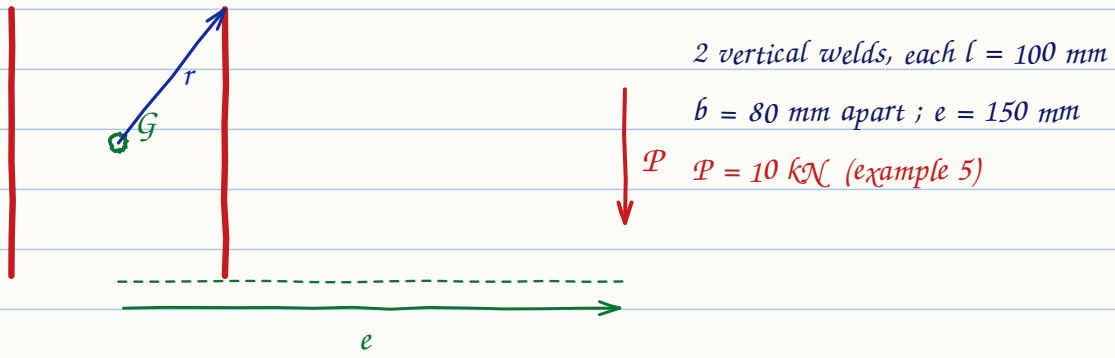


Fig 6: Eccentrically loaded weld group

$$\tau' = P / A = P / (0.707\, s\, L) \quad \text{(primary shear)}$$
$$\tau'' = T\, r / J = P\, e\, r / J \quad , \quad J = 0.707\, s\, J_u$$
$$J_u = \Sigma\, (l_i^3/12 + l_i\, r_i^2) \quad \text{(about } G\text{)}$$
$$\tau = \sqrt{(\tau'^2 + \tau''^2 + 2\, \tau'\, \tau''\, \cos\theta)} \le \tau_{allow}$$

- Locate centroid G : $\bar{x} = \Sigma\, l\, x / \Sigma\, l$, $\bar{y} = \Sigma\, l\, y / \Sigma\, l$; r_{max} = farthest weld end from G ; τ'' acts $\perp$ r.

Example 5 : (Fig 6) P = 10 kN, e = 150 mm, 2 welds l = 100, b = 80 : $A = 0.707\, s \times 200$ $\rightarrow \tau' = 70.7/s$. $J_u = 2\,[100^3/12 + 100 \times 40^2] = 4.87 \times 10^5$ mm³ $\rightarrow J = 344 \times 10^3\, s$. $T = 1.5 \times 10^6$ N·mm ; $r = \sqrt{(50^2 + 40^2)} = 64$ mm. $\tau'' = 1.5 \times 10^6 \times 64 / (344 \times 10^3\, s) = 279/s$. Components at the worst end : horizontal $279\,(50/64) = 218/s$, vertical $70.7 + 279\,(40/64) = 245/s$ $\rightarrow \tau = 328/s$.

τ_{allow} = 80 MPa → s = 328/80 = 4.1 mm → s = 5 mm (standard fillet size) ; minimum size also governed by plate thickness (≥ 3 mm for $t \le 6$).

Example 7 (bending) : bracket plate welded to a column by two horizontal fillet welds b = 100 mm long, placed d = 150 mm apart ; P = 8 kN at e = 200 mm. $M = 1.6\times10^6$ N·mm ; $Z_u = b\ d = 15000$ mm². $\sigma_b = M/(0.707\ s\ Z_u) = 151/s$. Direct shear $\tau' = 8000/(0.707\ s \times 200) = 56.6/s$. Resultant = $\sqrt{(151^2 + 56.6^2)}/s = 161/s$. With τ_{allow} = 80 → s = 2 mm ; but minimum practical size (thicker plate 10 mm → 5 mm) → s = 5 mm.

(c) Torsion of circular / welded tubular joints

Circular hollow shaft welded all round to a plate and twisted : $\tau = T\ d / 2J = 2T / (\pi\ d^2\ t_w)$ with $t_w = 0.707\ s$; combined with bending M : $\tau = \sqrt{(\tau_T^2 + \tau_b^2)}$, $\tau_b = 4M/(\pi\ d^2\ t_w)$.

Welding defects & comparison

- Defects : porosity (gas), slag inclusion, lack of fusion / penetration, undercut, cracks (hot / cold), overlap, distortion, residual stress → reduce static & fatigue strength ; detected by DPT, MPI, radiography, ultrasonic.

Point	Bolted	Welded
Permanence	detachable	permanent
Weight / cost	heavier (holes, plates, nuts)	lighter, cheaper
Fatigue	good with preload	poor at weld toe
Stress concentration	holes + threads	weld toe, residual stress
Skill / inspection	simple	skilled labour + NDT

Remember: Welded joints under fatigue are limited by weld toe : use stress range ($\Delta\sigma$) and K_f ; peen, grind toes, use full penetration welds to improve fatigue life. Weld size should not exceed the thinner plate thickness ; leg size ≤ 1.2 t, minimum 3-5 mm.

Topic	Key point
Preload	$F_i = 2840\ d$; $T = 0.2\ F_i\ d$
Joint constant	$C = k_b/(k_b+k_m)$; $F_b = F_i + C\ P$
Separation	$P_o = F_i/(1 - C)$
Eccentric (shear)	$F' = P/n$, $F'' = P\ e\ r/\Sigma r^2$, vector sum
Fillet weld	$P = 0.707\ s\ l\ \tau$ (×2 for double)
Eccentric weld	$\tau' = P/A$, $\tau'' = P\ e\ r/J$, $J = 0.707\ s\ J_u$